

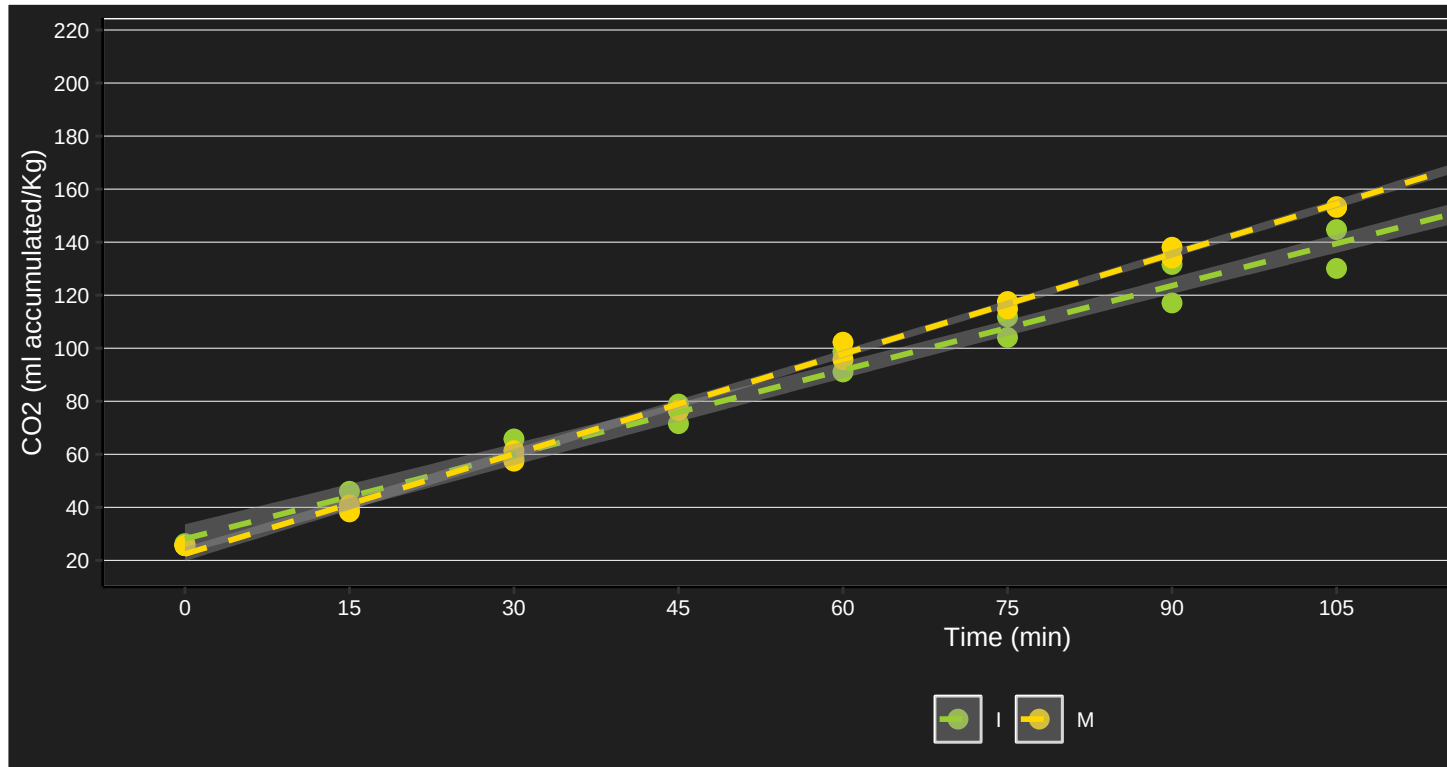
Ensayo 1

Respiración acumulada en frutos I y M



Immature and mature ubajay fruits were selected and randomly distributed in 4 jars, 2 immature and 2 mature, then respiration was quantified from accumulated CO₂ every 15 minutes for 150 minutes.

CO₂ acumulation

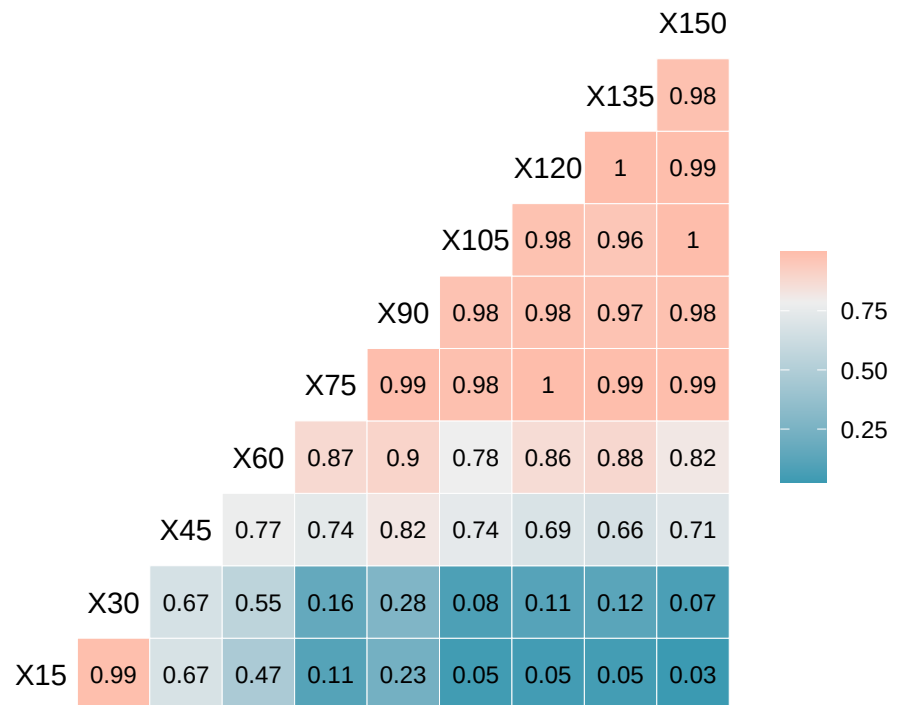


Descriptive table

```
## # A tibble: 20 x 7
## # Groups:   time_min [10]
##   time_min matu carbon_ac_n carbon_ac_Mean carbon_ac_sd carbon_ac_min
##   <fct>     <fct>      <int>      <dbl>      <dbl>      <dbl>
## 1 15       I           2        42.5        4.97       39.0
## 2 15       M           2        39.6        1.86       38.3
## 3 30       I           2        62.2        5.13       58.5
## 4 30       M           2        59.4        2.79       57.4
## 5 45       I           2        75.2        5.23       71.5
## 6 45       M           2        76.6        0.110      76.6
## 7 60       I           2        94.9        5.39       91.1
## 8 60       M           2        99.0        4.66       95.7
## 9 75       I           2       108.         5.50      104.
## 10 75      M           2       116.         1.97      115.
## 11 90       I           2       124.        10.3      117.
## 12 90       M           2       136.         2.91      134.
## 13 105      I           2       137.        10.4      130.
## 14 105      M           2       153.         0.221     153.
## 15 120      I           2       157.        10.5      150.
```

```
## 16 120      M      2      176.      4.77      172.
## 17 135      I      2      170.     10.6      163.
## 18 135      M      2      190.      6.59      185.
## 19 150      I      2      186.     15.4      176.
## 20 150      M      2      213.      3.02      211.
## # i 1 more variable: carbon_ac_max <dbl>
```

Correlations over time



Covariance matrix

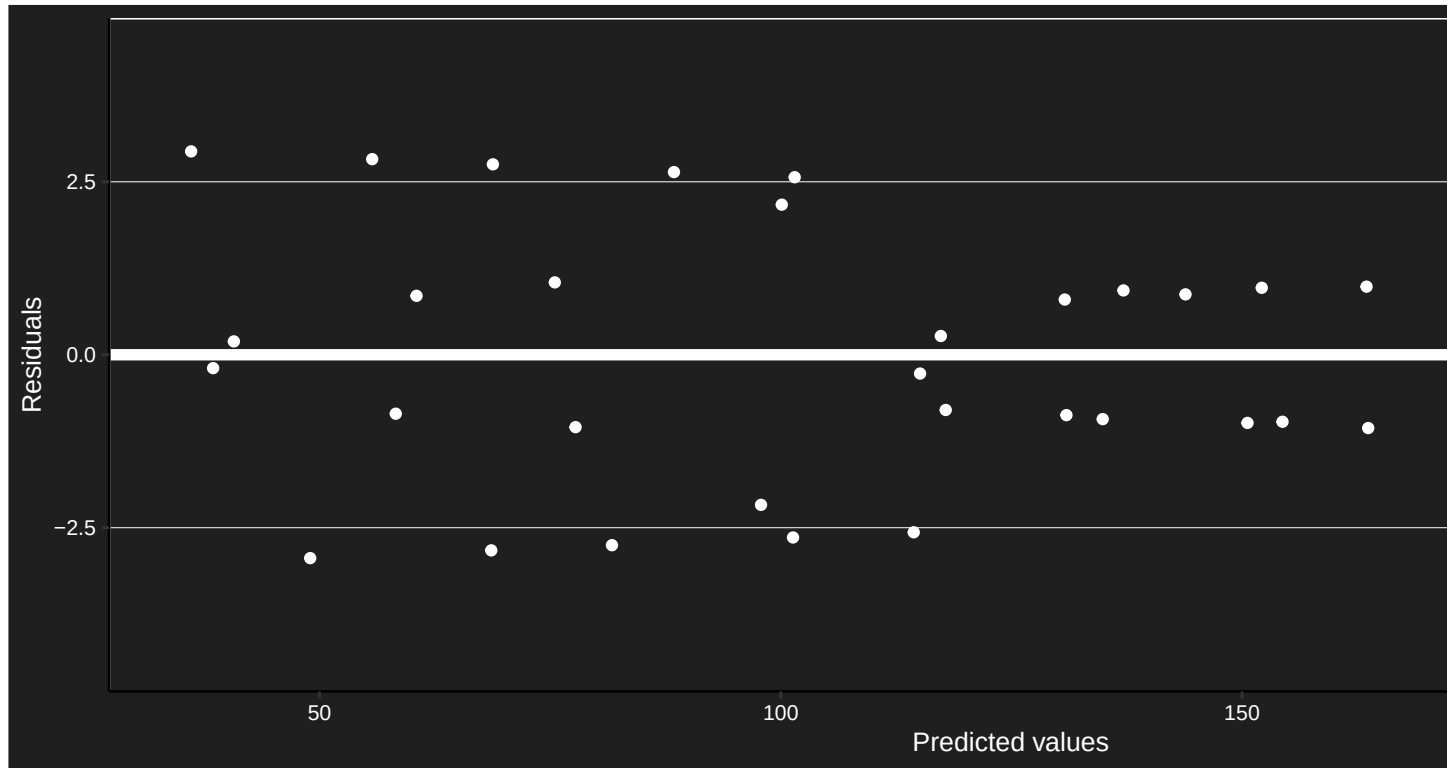
```
##      15      30      45      60      75      90      105      120      135      150
## 15 12.28 12.95  7.38  7.79   2.21   7.31   1.73   2.14   2.50   1.66
## 30 12.95 13.93  7.77  9.75   3.59   9.42   3.25   5.23   6.22   4.90
## 45  7.38  7.77  9.77 11.46  13.46  23.37  25.37  27.07  27.76  38.97
## 60  7.79  9.75 11.46 22.55  24.26  38.94  40.66  51.74  56.09  68.14
## 75  2.21  3.59 13.46 24.26  34.14  52.90  62.77  73.57  77.63 102.21
## 90  7.31  9.42 23.37 38.94  52.90  83.40  97.35 112.92 118.82 157.38
## 105 1.73  3.25 25.37 40.66  62.77  97.35 119.47 134.76 140.36 191.45
## 120 2.14  5.23 27.07 51.74  73.57 112.92 134.76 159.43 168.68 220.61
## 135 2.50  6.22 27.76 56.09  77.63 118.82 140.36 168.68 179.37 231.42
## 150 1.66  4.90 38.97 68.14 102.21 157.38 191.45 220.61 231.42 309.86
```

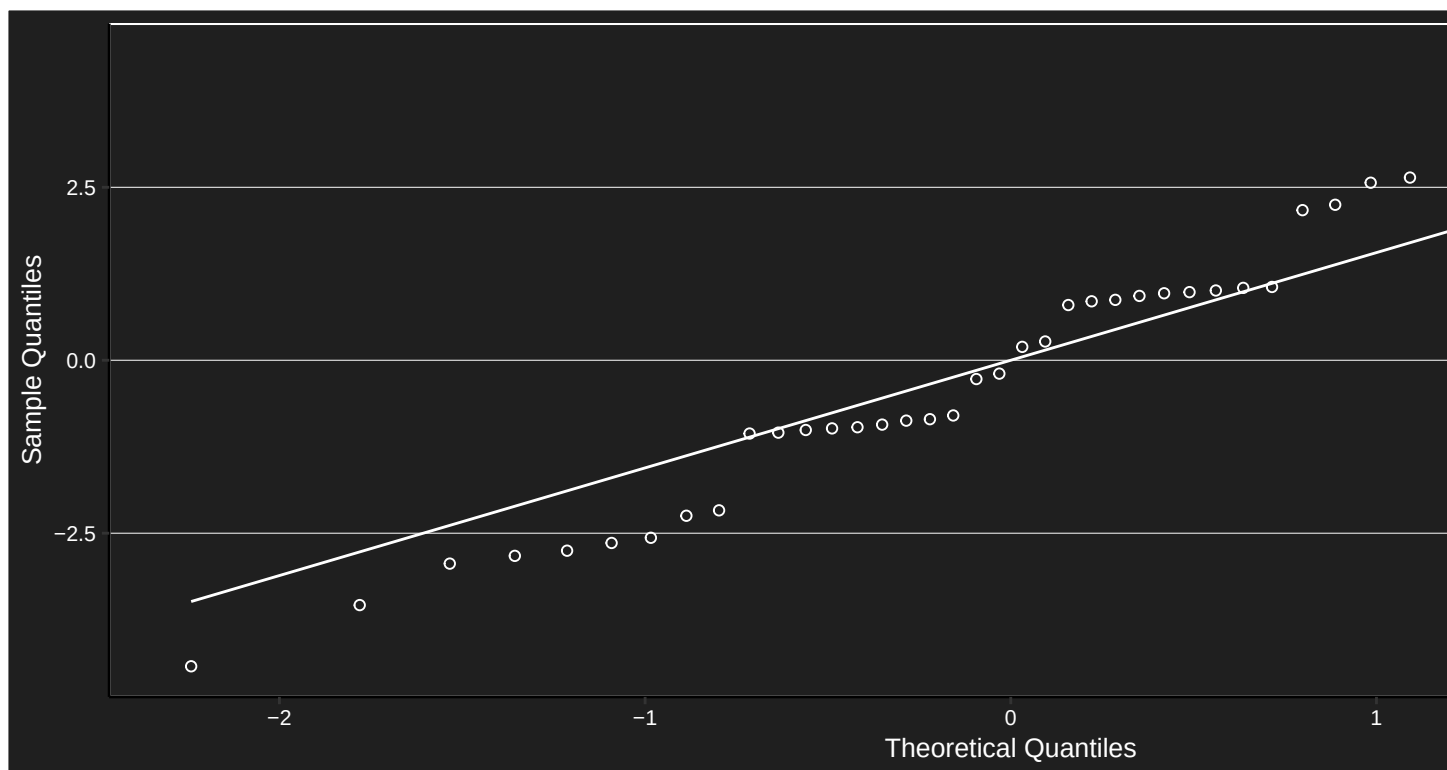
Marginal model with first-order autoregressive structure

```
## gls(model = (carbon_ac) ~ time_min * matu + basal, data = resp2w,
```

```
## correlation = corAR1(form = ~1 | rep))
```

Assumptions





```
##
##  Shapiro-Wilk normality test
##
## data:  e
## W = 0.97616, p-value = 0.5498
```

Model coefficients

```
##      (Intercept)      time_min30      time_min45      time_min60
## -1088.3098591      19.6250543      32.7084238      52.3334782
##      time_min75      time_min90      time_min105      time_min120
##      65.4168477      81.7897236      94.8730931      114.4981474
##      time_min135      time_min150      matuM      basal
##      127.5815169      143.9543928      24.0203019      43.2170578
## time_min30:matuM time_min45:matuM time_min60:matuM time_min75:matuM
##      0.1712202      4.3272497      7.0553453      11.2113748
## time_min90:matuM time_min105:matuM time_min120:matuM time_min135:matuM
##      14.6347734      18.7908030      21.5188986      22.4853372
## time_min150:matuM
##      29.0983267
```

Anova

```
## Denom. DF: 19
##      numDF  F-value p-value
## (Intercept)      1 12422.598 <.0001
## time_min      9  497.680 <.0001
```

```
## matu          1      23.524  0.0001
## basal         1      18.306  0.0004
## time_min:matu  9       4.645  0.0024
```

Simple effects

```
## $emmeans
## time_min = 15:
## matu      emmean      SE    df  lower.CL  upper.CL
## I      29.05593  3.996413  4.38   18.32747   39.78439
## M      53.07623  3.996413  4.26   42.24267   63.90980
##
## time_min = 30:
## matu      emmean      SE    df  lower.CL  upper.CL
## I      48.68099  3.996413  4.37   37.94735   59.41462
## M      72.87251  3.996413  4.38   62.14744   83.59757
##
## time_min = 45:
## matu      emmean      SE    df  lower.CL  upper.CL
## I      61.76435  3.996413  4.23   50.90664   72.62207
## M      90.11191  3.996413  4.30   79.31519  100.90863
##
## time_min = 60:
## matu      emmean      SE    df  lower.CL  upper.CL
## I      81.38941  3.996413  4.35   70.64128   92.13754
## M     112.46506  3.996413  4.37  101.73142  123.19869
##
## time_min = 75:
## matu      emmean      SE    df  lower.CL  upper.CL
## I      94.47278  3.996413  4.28   83.65659  105.28897
## M     129.70446  3.996413  4.31  118.91264  140.49627
##
## time_min = 90:
## matu      emmean      SE    df  lower.CL  upper.CL
## I     110.84565  3.996413  4.38  100.11719  121.57412
## M     149.50073  3.996413  4.26  138.66717  160.33429
##
## time_min = 105:
## matu      emmean      SE    df  lower.CL  upper.CL
## I     123.92902  3.996413  4.37  113.19539  134.66266
## M     166.74013  3.996413  4.38  156.01506  177.46520
##
## time_min = 120:
## matu      emmean      SE    df  lower.CL  upper.CL
## I     143.55408  3.996413  4.23  132.69637  154.41179
## M     189.09328  3.996413  4.30  178.29656  199.89000
##
## time_min = 135:
## matu      emmean      SE    df  lower.CL  upper.CL
## I     156.63745  3.996413  4.35  145.88932  167.38558
## M     203.14309  3.996413  4.37  192.40945  213.87672
##
## time_min = 150:
```

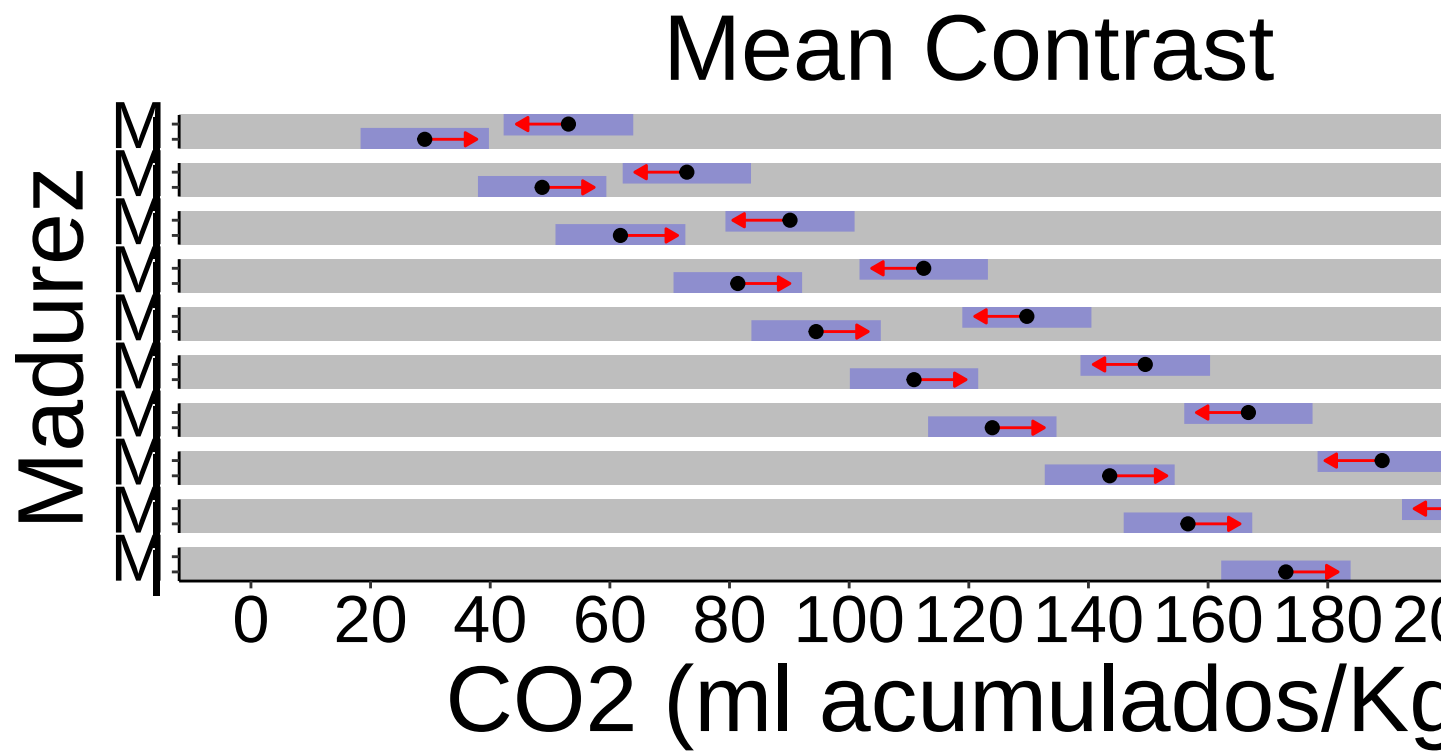
```

##   matu      emmean      SE    df lower.CL upper.CL
## I    173.01032 3.996413 4.28 162.19413 183.82652
## M    226.12895 3.996413 4.31 215.33714 236.92077
##
## Degrees-of-freedom method: satterthwaite
## Results are given on the ( (not the response) scale.
## Confidence level used: 0.95
##
## $contrasts
## time_min = 15:
## contrast estimate      SE    df t.ratio p.value
## I - M    -24.02030 7.197624 6.29  -3.337  0.0146
##
## time_min = 30:
## contrast estimate      SE    df t.ratio p.value
## I - M    -24.19152 7.197624 6.30  -3.361  0.0141
##
## time_min = 45:
## contrast estimate      SE    df t.ratio p.value
## I - M    -28.34755 7.197624 4.51  -3.938  0.0135
##
## time_min = 60:
## contrast estimate      SE    df t.ratio p.value
## I - M    -31.07565 7.197624 6.28  -4.317  0.0045
##
## time_min = 75:
## contrast estimate      SE    df t.ratio p.value
## I - M    -35.23168 7.197624 6.24  -4.895  0.0024
##
## time_min = 90:
## contrast estimate      SE    df t.ratio p.value
## I - M    -38.65508 7.197624 6.29  -5.371  0.0015
##
## time_min = 105:
## contrast estimate      SE    df t.ratio p.value
## I - M    -42.81110 7.197624 6.30  -5.948  0.0008
##
## time_min = 120:
## contrast estimate      SE    df t.ratio p.value
## I - M    -45.53920 7.197624 4.51  -6.327  0.0021
##
## time_min = 135:
## contrast estimate      SE    df t.ratio p.value
## I - M    -46.50564 7.197624 6.28  -6.461  0.0005
##
## time_min = 150:
## contrast estimate      SE    df t.ratio p.value
## I - M    -53.11863 7.197624 6.24  -7.380  0.0003
##
## Note: contrasts are still on the ( scale
## Degrees-of-freedom method: satterthwaite

```

Statistically significant differences were found in the CO₂ respiration rate in each time between immature and mature *Hexachlamys edulis* fruits.

Comparison chart



Fitted model plot

